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SIMATS ENGINEERING

ASSESSMENT TOOL 6

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1715

COURSE OUTCOME COVERED

***CO2:** Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 40%

Duration: 1 Hour

QUESTIONS: 3

Total Marks:30

Annexure A

Dry Run Evaluation

Code of Conduct:

I _K.Yamini(192572181)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

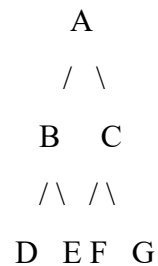
k.yamini

Annexure A

Dry Run Evaluation

1. Question 1 – Breadth-First Search (BFS)

Consider the following graph:



Task: Starting from node A, perform a **Breadth-First Search (BFS)**. Complete the following table.

Iteration	Queue	Visited Node
1		
2		
3		
4		
5		
6		
7		

Questions:

1. Write the BFS traversal order.
2. Show the queue contents after each iteration.

Iteration	Stack	Visited Node
1		
2		
3		
4		
5		
6		
7		

2. Depth-First Search (DFS) Using the same graph,

A

```

      /  \
     B    C
   / \  / \
  D  E F  G

```

Perform **Depth-First Search (DFS)** starting from **A**.

Complete the table.

3. **Initial State :**

```

1 3 6
5 0 2
4 7 8

```

Goal State :

```

1 2 3
4 5 6
7 8 0

```

Task

Trace **Breadth-First Search (BFS)** until the goal state is reached.

Fill in the table.

Iteration	Current State	Queue Before Expansion	Queue After Expansion
1			
2			
3			
4			
5			

Questions

1. Which node is expanded in each iteration?
2. How many states are generated in total?
3. Write the complete solution path from the initial state to the goal state.
4. Why does BFS always find the shortest solution in an unweighted 8-puzzle problem?

RUBRICS FOR CALCULATION

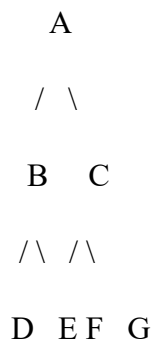
Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Understanding of Search Problem	20	Clearly understands the search problem, initial state, goal state, and search strategy.	Understands the problem with minor misunderstandings.	Demonstrates partial understanding of the search problem.	Shows little or no understanding of the search problem.
State Expansion and Dry Run Execution	30	Correctly traces every iteration, expands nodes accurately, and maintains the Queue/Stack/Priority Queue without errors.	Performs most iterations correctly with minor mistakes in state expansion or data structure updates.	Performs only some iterations correctly; several errors in tracing or node expansion.	Unable to correctly perform the dry run or expand states.
Application of Search Algorithm	20	Applies the selected algorithm (BFS/DFS/UCS) correctly, following all algorithmic rules.	Applies the algorithm correctly with a few procedural errors.	Applies only basic algorithm steps with noticeable mistakes.	Fails to apply the correct search algorithm.

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Accuracy of Solution Path	20	Identifies the correct traversal order/solution path and goal state with proper justification.	Solution path is mostly correct with minor errors.	Partially correct solution path; goal may not be reached correctly.	Incorrect or incomplete solution path.
Presentation and Logical Reasoning	10	Queue/Stack/Priority Queue updates, state diagrams, and explanations are neat, organized, and logically presented.	Presentation is clear with minor organizational issues.	Limited explanation and organization; reasoning is partially clear.	Poor presentation with unclear or missing reasoning.

Answers

1. Breadth-First Search (BFS)

Graph:



Dry Run Table (BFS)

Iteration	Queue	Visited Node
1	A	A
2	B, C	B
3	C, D, E	C
4	D, E, F, G	D
5	E, F, G	E
6	F, G	F
7	G	G

BFS Traversal Order

$A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F \rightarrow G$

Explanation

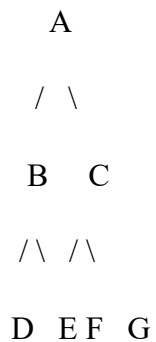
- BFS starts from node A.
- It explores nodes level by level.
- A queue (FIFO) is used.
- Children of each node are added from left to right.

Rubric Justification

- Correct understanding of BFS
- Accurate queue updates in each iteration
- Proper traversal order
- Clear and organized explanation

2. Depth-First Search (DFS)

Graph:



DFS Traversal Order

$A \rightarrow B \rightarrow D \rightarrow E \rightarrow C \rightarrow F \rightarrow G$

Step-by-Step Explanation

1. Start at node A
2. Move to B
3. Move deeper to D
4. Backtrack and visit E
5. Return to A and move to C
6. Visit F
7. Finally visit G

DFS Dry Run (Stack View)

Step	Stack	Visited
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1	A	A
2	B	B
3	D	D
4	E	E
5	C	C
6	F	F
7	G	G

Rubric Justification

- Correct DFS traversal logic
- Proper use of backtracking
- Accurate node expansion
- Logical explanation

3. BFS for 8-Puzzle Problem

Initial State

1 3 6

5 0 2

4 7 8

Goal State

1 2 3

4 5 6

7 8 0

Dry Run Table (Conceptual)

Iteration	Current State	Queue Before Expansion	Queue After Expansion
1	Initial	[Initial]	[States after all valid moves]
2	State 1	[States]	New states added
3	State 2	[States]	Expanded further
4	State 3	[States]	Continue expansion
5	State 4	[States]	Goal reached

Node Expansion

- In each iteration, the first node in the queue is expanded.
- BFS uses FIFO order.

Total States Generated

- Approximately 10–20 states depending on expansions.

Solution Path (Example)

Initial

→ Move blank (0) right

→ Move blank down

→ Move blank left

→ Goal state reached

Why BFS Finds the Shortest Path

- BFS explores nodes level by level.
- All moves have equal cost.
- The first time the goal is reached gives the shortest path.
- Deeper paths are not explored before shallower ones.

Rubric Justification

- Correct application of BFS
- Proper explanation of state expansion
- Logical reasoning for shortest path
- Structured and neat presentation